

MPG No. 8.1.24 *mmm* [Type I, orthorhombic] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & M_{yzx} \end{bmatrix}$$

$$M_{yzx} = \mathbb{M}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & M_{zxy} & 0 \\ 0 & 0 & -M_{yzx} - M_{zxy} \end{bmatrix}$$

$$M_{yzx} = 3\mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$

$$M_{zxy} = -3\mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & 0 \\ 0 & M_{zxy} & 0 \\ 0 & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)} + \mathbb{T}_v^{(2)}$$

$$M_{zxy} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)} - \mathbb{T}_v^{(2)}$$

$$M_{xyz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).